Report 4.10 – Risultati per k con $TIMES = 10$

2 maggio 2026

1 Grafici dei risultati per k e $TIMES = 10$

I grafici mostrano rispettivamente l'andamento di:

- Avg fQ
- fQ Min Found
- Profit Found
- Weight Found
- Profit GAP
- Weight GAP
- n of fQ Min Found

al variare di $k \in \{1000, 2000, 3000, 4000, 5000\}$, confrontando i risultati usando QALS e NON utilizzando QALS.

I risultati riportati sono stati ottenuti utilizzando:

- $\lambda = \frac{\sum_{i=1}^{n} p_i}{650 \cdot C}$
- l'istanza `ksp_3.txt`

Profit ottimo	2679
Weight ottimo	496
TIMES	10

Tabella 1: Soluzione di riferimento del problema di knapsack.

2 Tabella riassuntiva

k	QALS							NO QALS						
	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min	Avg fQ	fQ Min	Profit	Weight	Profit GAP	Weight GAP	n fQ Min
1000	152357.55	84302.41	2662	2390	17	-1894	1	-9424.8	-9424.8	2936	560	-257	-64	10
2000	155488.02	118747.24	3872	2721	-1193	-2225	1	-9424.8	-9424.8	2936	560	-257	-64	10
3000	148682.71	33336.15	2433	1772	246	-1276	1	-9424.8	-9424.8	2936	560	-257	-64	10
4000	162673.57	111936.55	4701	2669	-2022	-2173	1	-9424.8	-9424.8	2936	560	-257	-64	10
5000	179905.33	113582.99	3282	2671	-603	-2175	1	-9424.8	-9424.8	2936	560	-257	-64	10

Tabella 2: Risultati numerici per $k \in \{1000, 2000, 3000, 4000, 5000\}$ con `TIMES = 10`.

3 Grafici

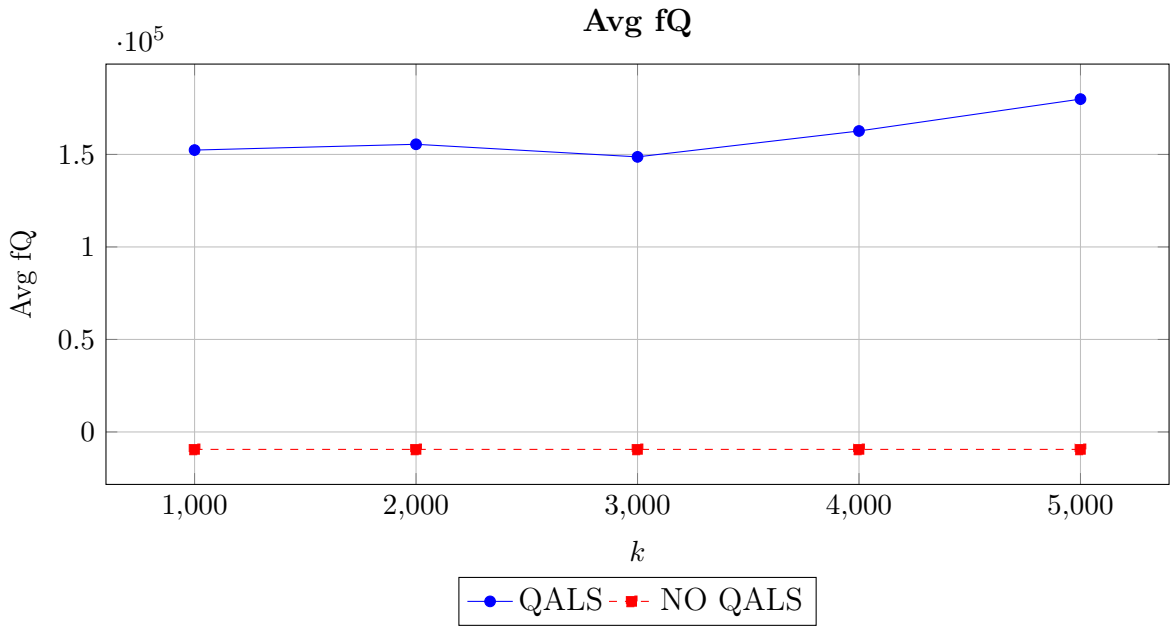

Figura 1: Andamento di Avg fQ al variare di k.

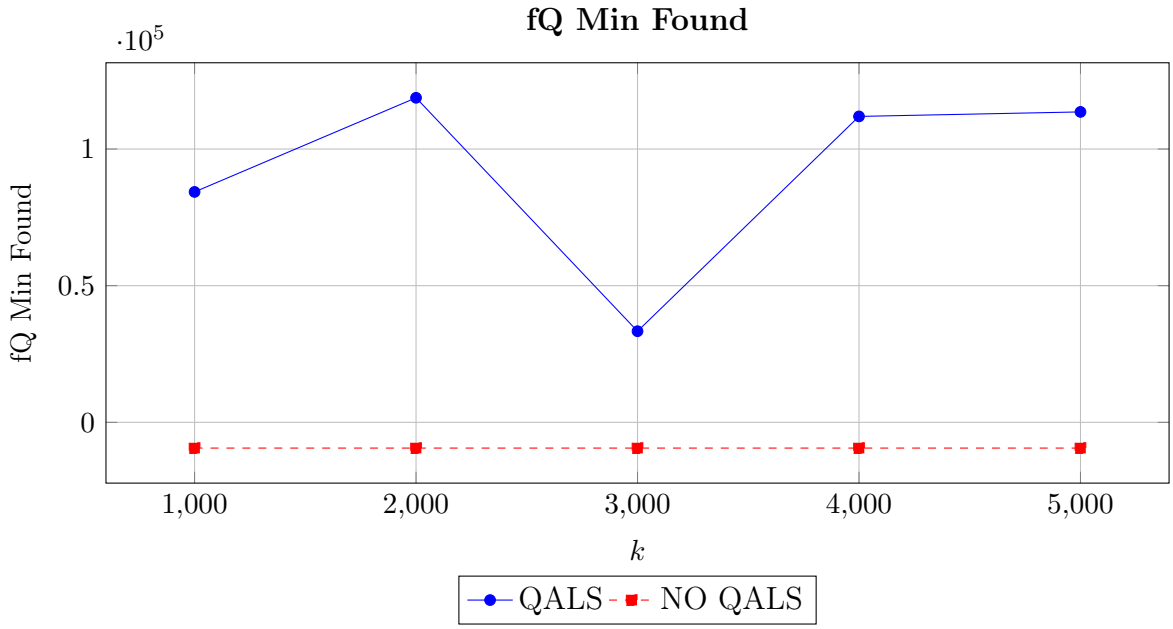

Figura 2: Andamento di fQ Min Found al variare di k.

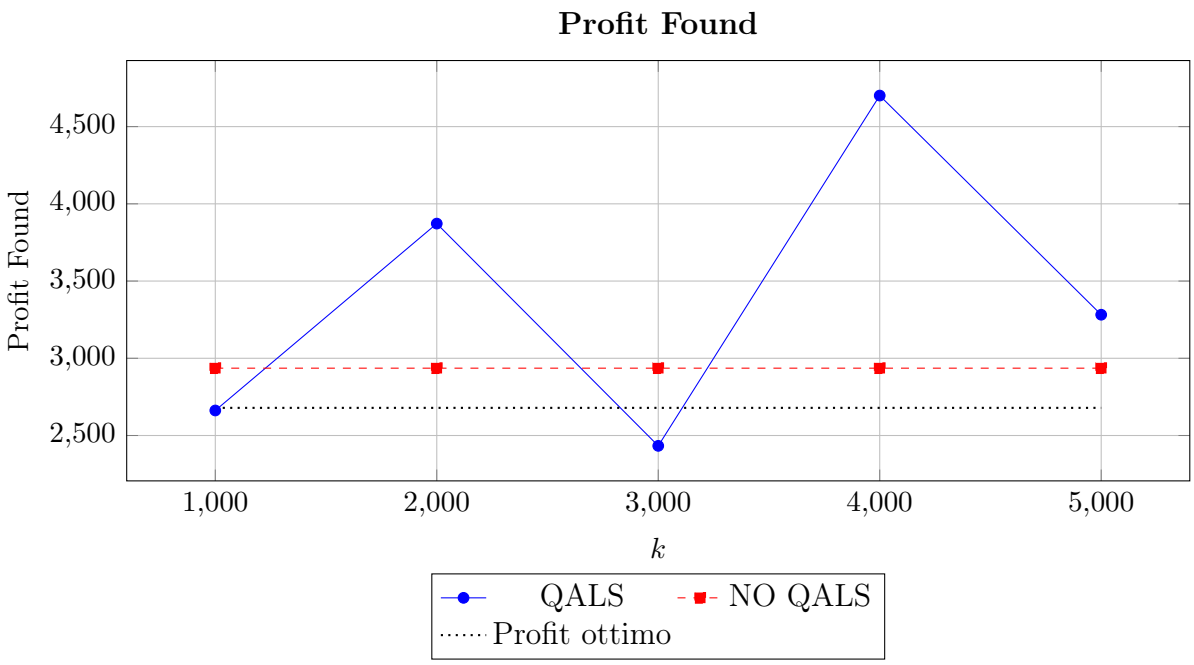

Figura 3: Andamento del profitto trovato al variare di k.

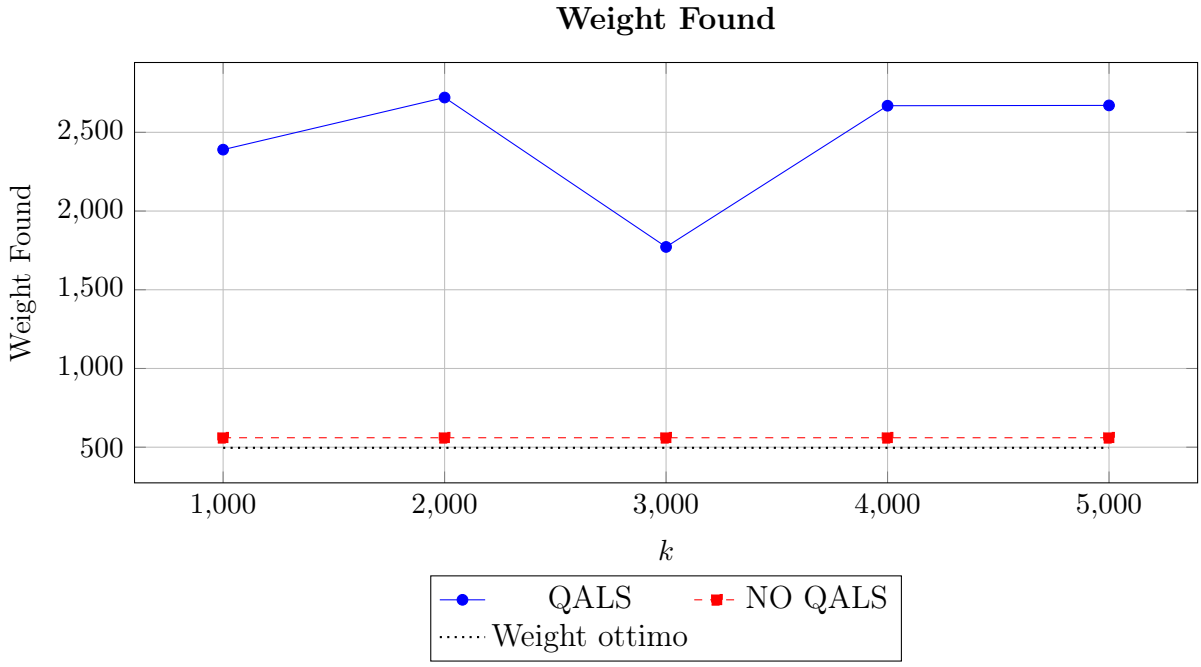

Figura 4: Andamento del peso trovato al variare di k.

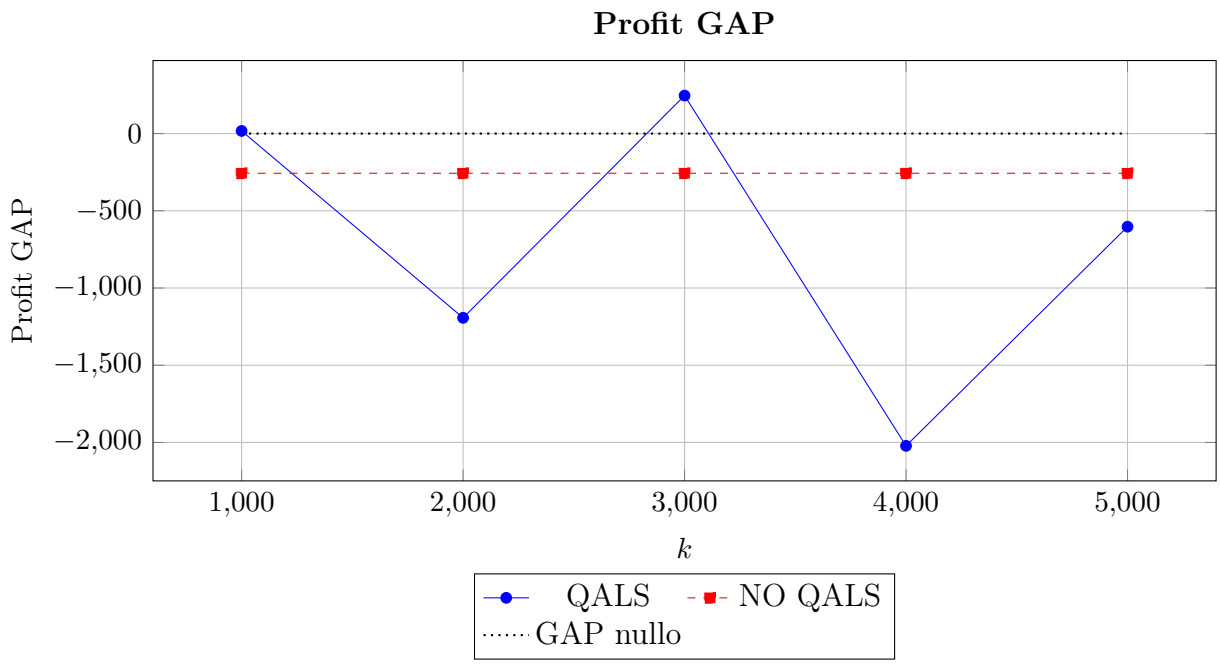

Figura 5: Andamento del Profit GAP al variare di k.

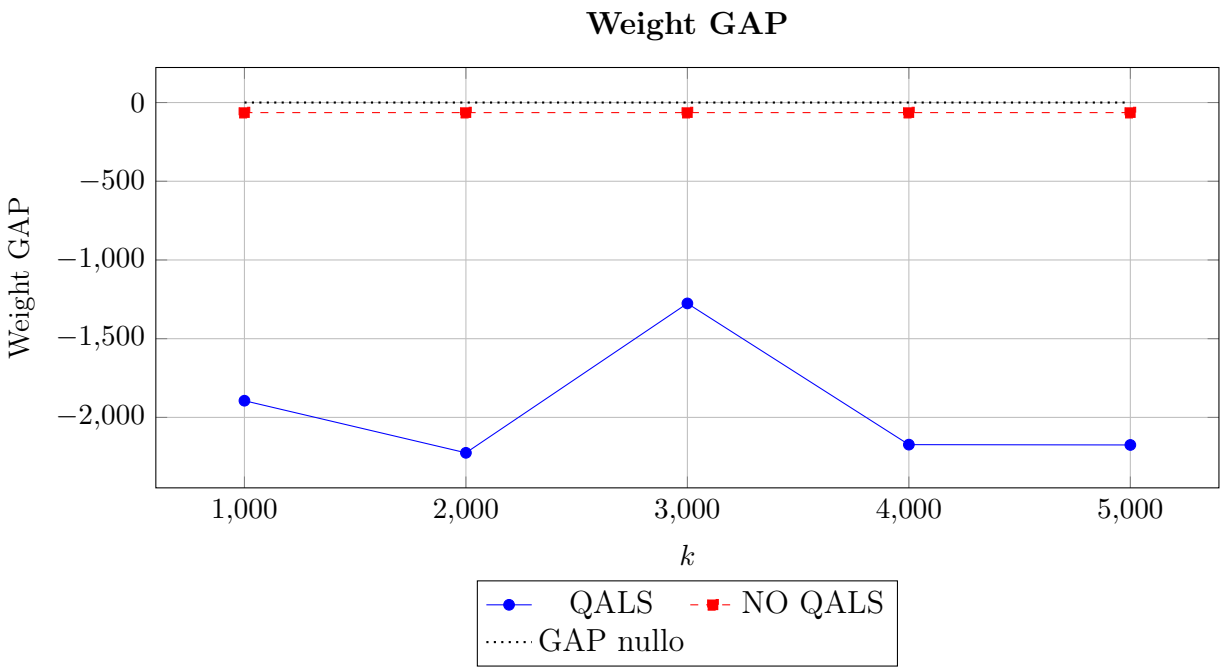

Figura 6: Andamento del Weight GAP al variare di k.

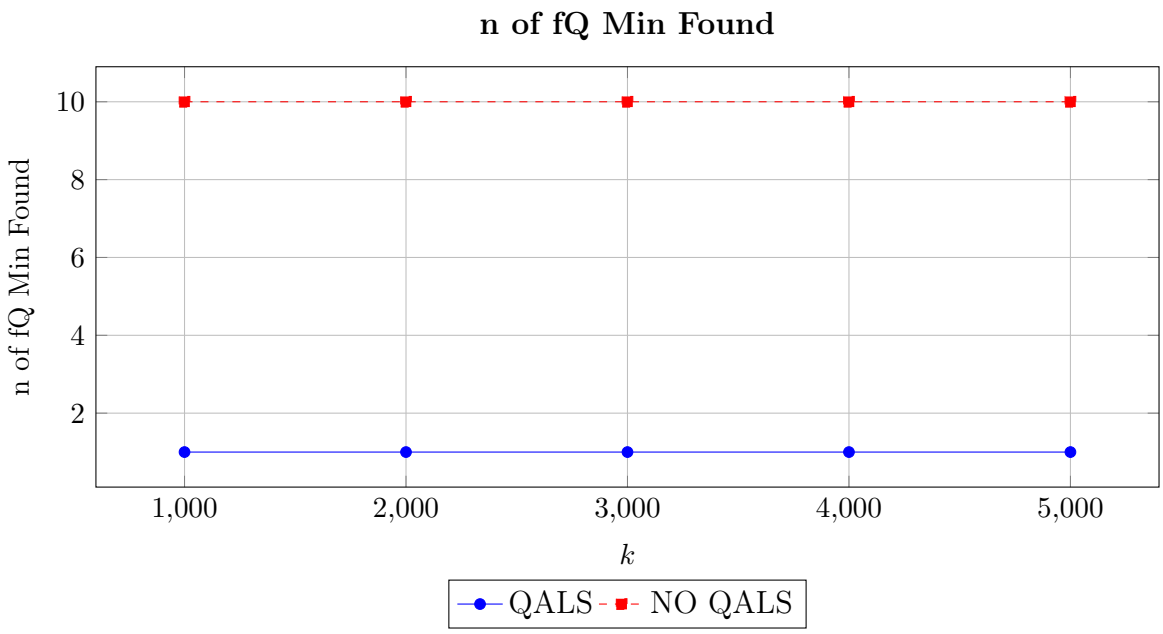

Figura 7: Numero di volte in cui è stato trovato il minimo fQ al variare di k.